Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese	**Department of Electrical, Electronic and Computer Engineering**
Eksamenvraestel	**Examination Question Paper**
Module EBN111	**Module EBN111**
Elektrisiteit en Elektronika	**Electricity and Electronics**
9 Junie 2008	9 June 2008

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
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Eksameninligting: *Examination information:*			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	3 ure *3 hours*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		21	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Hierdie vraestel mag slegs in Afrikaans of Engels beantwoord word.
   This question paper may only be answered in Afrikaans or English.

Interne Eksaminator(e): *Internal examiner(s):*	1 Prof J Joubert 2 Mr JP de Villiers 3 Mr JH van Wyk 4 Mr E Kruger	Eksterne Eksaminator: *External Examiner:*	Prof JW Odendaal

VRAAG 1 / QUESTION 1

Vraag 1.1	Question 1.1
Vraag 1.1 verwys na Fig. Q1.1. In die stroombaan getoon sal 'n verlaging in R_3 lei tot 'n verlaging van: A) Die stroom deur R_3. B) Die spanning oor R_3. C) Die spanning oor R_1. D) V_s. E) Geen van bogenoemde nie.	Question 1.1 refers to Fig. Q1.1. In the circuit shown, a decrease in R_3 leads to a decrease of: A) The current through R_3. B) The voltage across R_3. C) The voltage across R_1. D) V_s. E) None of the above.

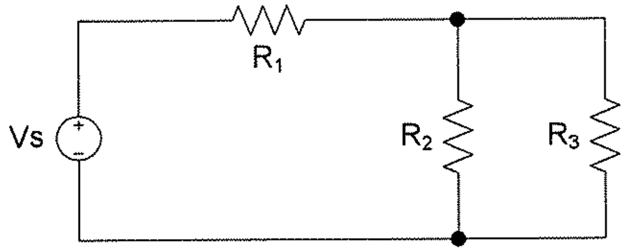


Fig. Q1.1

Vraag 1.2	Question 1.2
Vraag 1.2 verwys na Fig. Q1.2. Vir die stroombaan getoon, wat is die waarde van G_{eq}? A) 18 S B) 2.788 S C) 6 S D) 10 S E) Geen van bogenoemde nie.	Question 1.2 refers to Fig. Q1.2. For the circuit shown, what is the value of G_{eq}? A) 18 S B) 2.788 S C) 6 S D) 10 S E) None of the above.

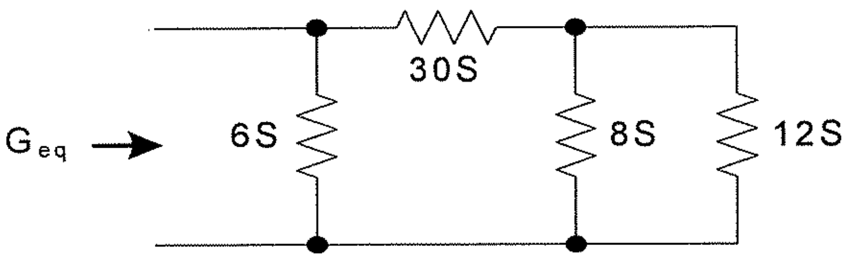


Fig. Q1.2

Vraag 1.3	Question 1.3
Vraag 1.3 verwys na Fig. Q1.3. Watter van die volgende stroombane volg nie Kirchhoff se Spanningswet nie?	Question 1.3 refers to Fig. Q1.3. Which of the following circuits does not follow Kirchoff's Voltage Law?

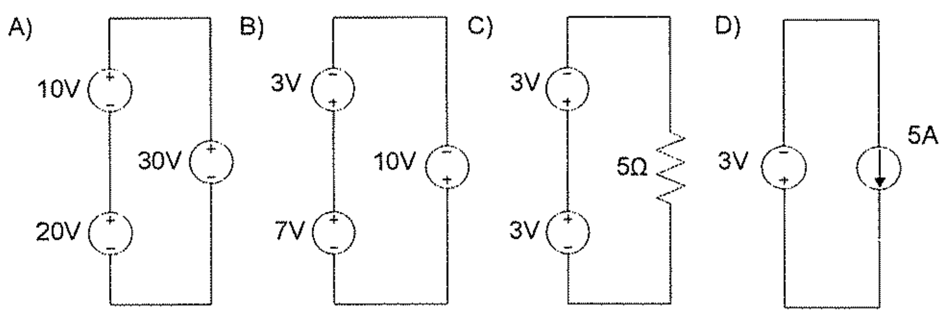


Fig. Q1.3

Vraag 1.4	Question 1.4
Wanneer 'n stroombron gedeel word deur twee kringe, vorm A) Een van die twee kringe 'n superkring. B) Beide die kringe superkringe. C) Die twee kringe en enige serie elemente saam 'n superkring. D) Die twee kringe en enige parallele elemente saam 'n superkring. E) Die twee kringe saam 'n superkring F) Geen van bogenoemde nie.	When a current source is shared by two meshes, then A) One of the two meshes forms a supermesh. B) Both meshes form supermeshes. C) The two meshes and any series elements together form a supermesh. D) The two meshes and any parallel elements together form a supermesh. E) The two meshes together form a supermesh. F) None of the above.

Vraag 1.5	Question 1.5
'n Bipolêre spervlaktransistor is 'n A) onafhanklike stroombeheerde stroombron B) afhanklike spanningsbeheerde stroombron C) afhanklike stroombeheerde spanningsbron D) afhanklike stroombeheerde stroombron E) onafhanklike stroombeheerde spanningsbron	A Bipolar junction transistor is a A) independent current controlled current source B) dependent voltage controlled current source C) dependent current controlled voltage source D) dependent current controlled current source E) independent current controlled voltage source

Vraag 1.6.	Question 1.6.
Vraag 1.6 verwys na Fig. Q1.6. Bepaal die Norton ekwivalente weerstand van die stroombaan. A) 1 Ω B) 2 Ω C) 4 Ω D) 6 Ω E) 3 Ω F) 5 Ω G) Geen van die bogenoemdes nie	Question 1.6 refers to Fig. Q1.6. Find the Norton equivalent resistance for the circuit. A) 1 Ω B) 2 Ω C) 4 Ω D) 6 Ω E) 3 Ω F) 5 Ω G) None of the above

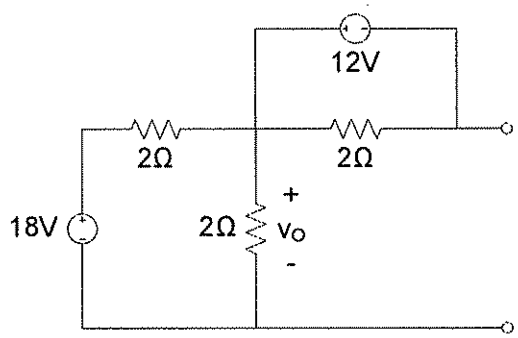


Fig. Q1.6

Vraag 1.7.	Question 1.7.
Vraag 1.7 verwys na Fig. Q1.7. As $R_f = 10$ kΩ, $R_1 = 40$ kΩ, $R_2 = 500$ kΩ, $R_3 = 250$ kΩ en $R_4 = 250$ kΩ is in die stroombaan, bepaal die analog uitset vir $V_1 = 1V$, $V_2 = 0V$, $V_3 = 1V$ en $V_4 = 0V$. A) -0.35 V B) 0.35 V C) -0.04 V D) 0.04 V E) -0.29 V F) 0.29 V G) Geen van die bogenoemdes nie	Question 1.7 refers to Fig. Q1.7. In the op-amp circuit, let $R_f = 10$ kΩ, $R_1 = 40$ kΩ, $R_2 = 500$ kΩ, $R_3 = 250$ kΩ and $R_4 = 250$ kΩ. Obtain the analogue output if $V_1 = 1V$, $V_2 = 0V$, $V_3 = 1V$ en $V_4 = 0V$. A) -0.35 V B) 0.35 V C) -0.04 V D) 0.04 V E) -0.29 V F) 0.29 V G) None of the above

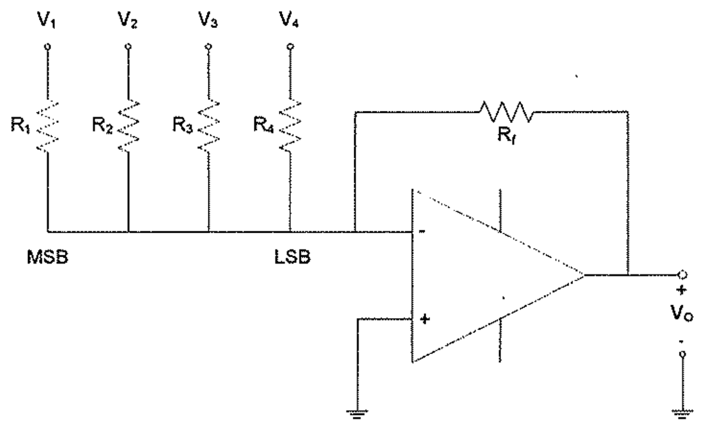


Fig. Q1.7

Vraag 1.8.	**Question 1.8.**
Vraag 1.8 verwys na Fig. Q1.8. Wat is die uitset spanning oombliklik nadat die skakelaar gesluit word? A) 0V B) 5V C) Op-amp afsny spanning D) Oneindig E) Geen van die bogenoemdes nie	Question 1.8 refers to Fig. Q1.8. Immediately after the switch is closed, the output voltage is: A) 0V B) -5V C) Op-amp saturation voltage D) Infinite E) None of the above

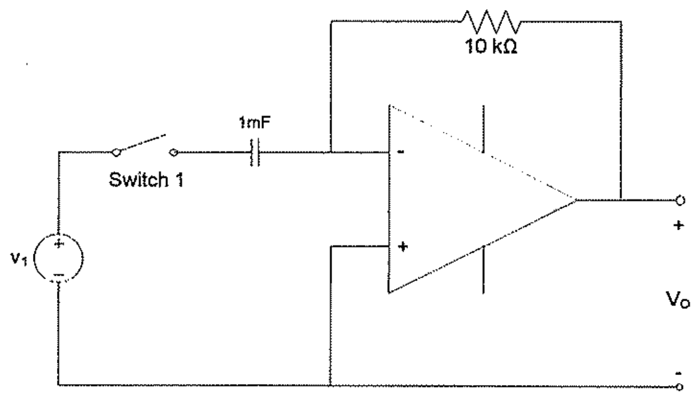


Fig. Q1.8

Vraag 1.9.	**Question 1.9.**
Vraag 1.9 verwys na Fig. Q1.9. Bepaal v_O. A) 0.8 V B) -3.2 V C) 6.4 V D) 3.2 V E) -6.4 V F) Geen van die bogenoemdes nie	Question 1.9 refers to Fig. Q1.9. Calculate v_O. A) 0.8 V B) -3.2 V C) 6.4 V D) 3.2 V E) -6.4 V F) None of the above

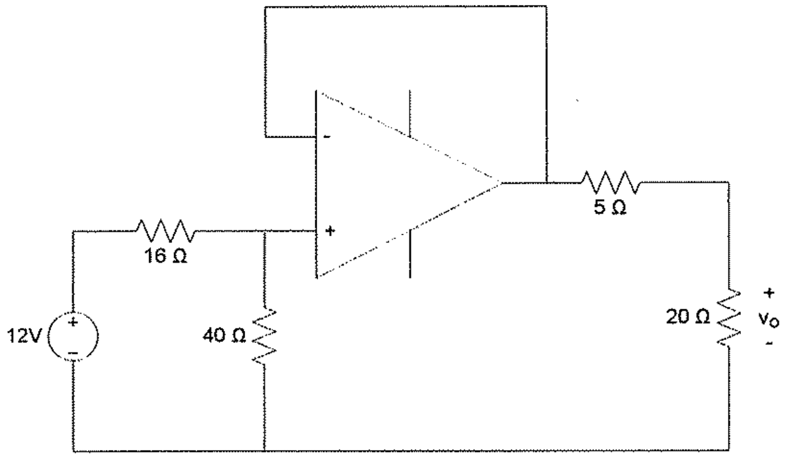


Fig. Q1.9

Vraag 1.10	**Question 1.10**
Bereken die ekwivalente kapasitansie tussen terminale *a* en *b* in fig. Q1.10. A) 4.42 μF B) 4.4 μF C) 4.53 μF D) 5 μF E) 17.69 μF F) 20 μF G) Geen een van bogenoemdes	Calculate the equivalent capacitance between terminals *a* and *b* in fig. Q1.10. A) 4.42 μF B) 4.4 μF C) 4.53 μF D) 5 μF E) 17.69 μF F) 20 μF G) None of the above

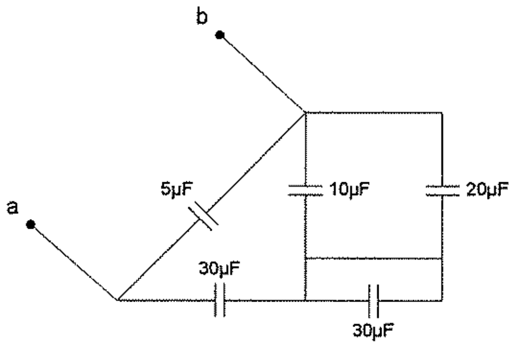


Fig. Q1.10

Vraag 1.11.	**Question 1.11.**
Vraag 1.11 verwys na Fig. Q1.11. Bepaal v_L. A) 12 V B) 10 V C) 8 V D) 15 V E) 6.4 V F) Geen van die bogenoemdes nie	Question 1.11 refers to Fig. Q1.11. Calculate v_L. A) 12 V B) 10 V C) 8 V D) 15 V E) 6.4 V F) None of the above

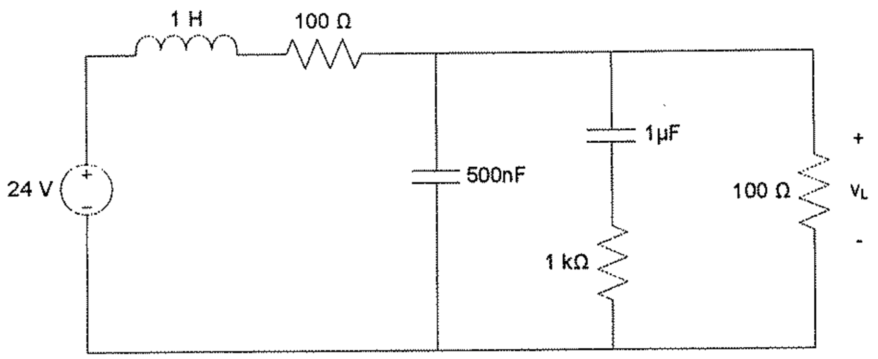


Fig. Q1.11

Vraag 1.12	**Question 1.12**
Die ooplus-versterking van 'n ideale operasionele versterker is: A) 0 B) Klein C) Baie groot D) Oneindig E) Afhanklik van die inset spannings F) Geen van die bogenoemdes nie	The open-loop gain of an ideal operational amplifier is: A) 0 B) Small C) Extremely large D) Infinite E) Dependant on input voltages F) None of the above

Vraag 1.13	**Question 1.13**
Vraag 1.13 verwys na Fig. Q1.13. Bepaal watter van die stroombane ekwivalent is. A) a en b B) b en d C) a en c D) c en d E) a en d F) Geen van die bogenoemdes nie	Question 1.13 refers to Fig. Q1.13. Determine which pair of circuits are equivalent. A) a and b B) b and d C) a and c D) c and d E) a and d F) None of the above

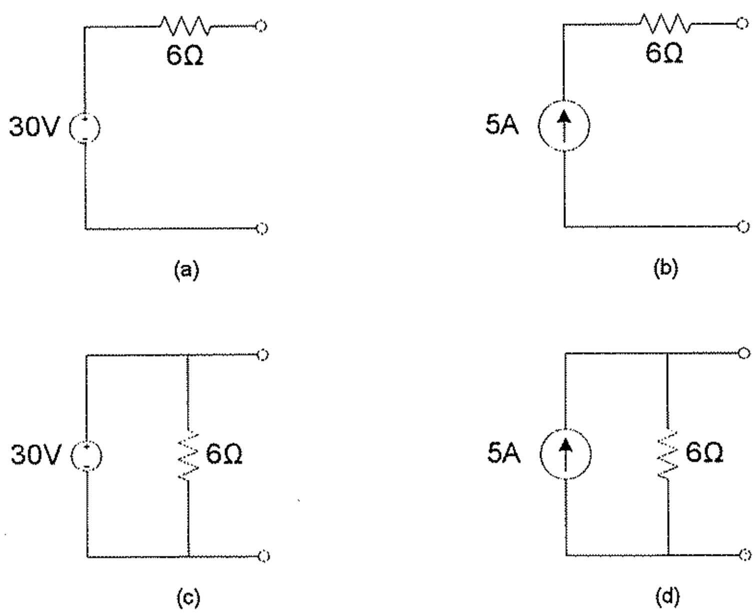


Fig. Q1.13

Vraag 1.14	**Question 1.14**
Vraag 1.14 verwys na Fig. Q1.14. Vind die maksimum drywing wat deur die stroombaan gelewer kan word. A) 0.125 W B) 0.18 W C) 0.333 W D) 0.75 W E) 1.08 W F) Geen van die bogenoemdes nie	Question 1.14 refers to Fig. Q1.14. Find the maximum power that can be delivered by the circuit. A) 0.125 W B) 0.18 W C) 0.333 W D) 0.75 W E) 1.08 W F) None of the above

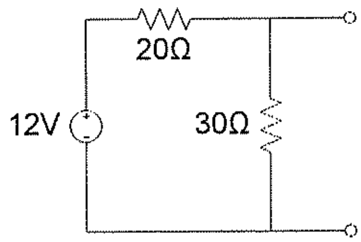


Fig. Q1.14

Vraag 1.15	**Question 1.15**
Vraag 1.15 verwys na Fig. Q1.15. Gebruik brontransformasie om die waarde van i_O te bepaal. A) 0.75A B) 1A C) 1.25A D) 2A E) 3A F) Geen van die bogenoemdes nie	Question 1.15 refers to Fig. Q1.15. Apply source transformation to determine i_O. A) 0.75A B) 1A C) 1.25A D) 2A E) 3A F) None of the above

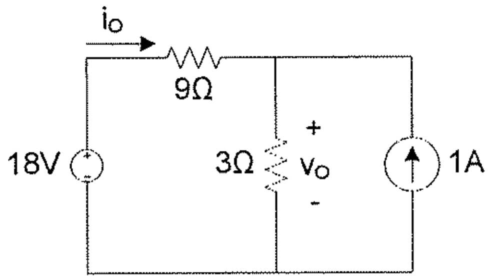


Fig. Q1.15

Vraag 1.16	**Question 1.16**
Vraag 1.16 verwys ook na Fig. Q1.15. Gebruik brontransformasie om die waarde van v_O te bepaal. A) 18V B) 6.75 C) 5V D) 15V E) 18.75V F) Geen van die bogenoemdes nie	Question 1.16 also refers to Fig. Q1.15. Apply source transformation to determine v_O. A) 18V B) 6.75 C) 5V D) 15V E) 18.75V F) None of the above

Vraag 1.17	Question 1.17
Die intree-weerstande van 'n ideale operasionele versterker is: A) Albei nul. B) Albei oneindig. C) Die omkeer inset is nul en die nie-omkeer inset is oneindig. D) Die omkeer inset is oneindig en die nie-omkeer inset is nul. E) Die omkeer inset is baie groot en die nie-omkeer inset is nul. F) Geen van die bogenoemdes nie.	The input resistances of an ideal operational amplifier are: A) Both zero. B) Both infinite. C) The inverting input is zero and the non-inverting input is infinite. D) The inverting input is infinite and the non-inverting input is zero. E) The inverting input is very large and the non-inverting input is zero. F) None of the above.

Vraag 1.18	Question 1.18
As 'n spanning van 5V op die nie-omkeer inset van 'n ideale operasionele versterker geplaas word, is die spanning op die omkeer inset: A) 0V B) 2.5V C) 5V D) -5V E) Oneindig. F) Geen van die bogenoemdes nie.	If the voltage applied to the non-inverting input of an ideal operational amplifier is 5V, the voltage at the inverting input is: A) 0V B) 2.5V C) 5V D) -5V E) Infinite. F) None of the above.

Vraag 1.19	Question 1.19
Bereken die ekwivalente induktansie tussen terminale *a* en *b* in fig. Q1.19. A) 1.71 H B) 3.53 H C) 10 H D) 10.46 H E) 34 H F) 35.8 H G) Geen van die bogenoemdes nie	Calculate the equivalent inductance between terminals *a* and *b* in fig. Q1.19. A) 1.71 H B) 3.53 H C) 10 H D) 10.46 H E) 34 H F) 35.8 H G) None of the above

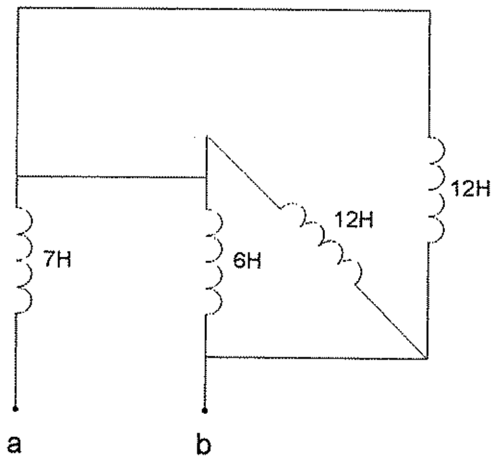


Fig. Q1.19

Vraag 1.20	**Question 1.20**
Watter een van die volgende stellings in verband met kapasitors is vals? A) ‘n Kapasitor bestaan uit twee geleidende plate wat geskei is deur ‘n diëlektrikum. B) ‘n Kapasitor is ‘n ope-baan vir DC (gelykstroom). C) Die spanning oor ‘n kapasitor kan nie oombliklik verander nie. D) Die ekwivalente kapasitansie van parallel gekonnekteerde kapasitors is die som van die individuele kapasitansies. E) Geen van die bogenoemdes nie	Which one of the statements regarding capacitors is false? A) A capacitor consists of two conducting plates separated by a dielectric. B) A capacitor is an open circuit to DC. C) The voltage over a capacitor can not change instantaneously. D) The equivalent capacitance of parallel-connected capacitors is the sum of the individual capacitances. E) None of the above.

Vraag 1.21	**Question 1.21**
Die resonante frekwensie (f_O) in ‘n stroombaan met ‘n enkele inductor en enkele kapsitor word gegee deur (1). In fig. Q1.21, bepaal die waarde van L_1 om ‘n resonante frekwensie van 1.0066 MHz te verkry. $f_O = \frac{1}{2\pi\sqrt{LC}}$ (1) A) 25 µH B) 2.5 µH C) 3.5 µH D) 10 µH E) 1 µH F) 50 µH G) Geen van die bogenoemdes nie	In a circuit with single inductor and capacitor, a frequency of resonance (f_O) is defined as in (1). In fig. Q1.21, find the value of L_1 necessary to make the circuit resonate at a frequency of 1.0066 MHz. $f_O = \frac{1}{2\pi\sqrt{LC}}$ (1) A) 25 µH B) 2.5 µH C) 3.5 µH D) 10 µH E) 1 µH F) 50 µH G) None of the above

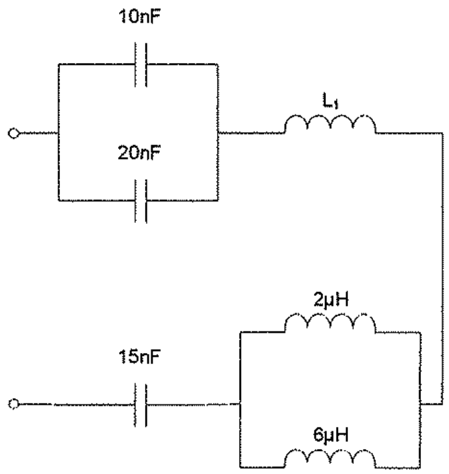


Fig. Q1.21

Vraag 1.22	**Question 1.22**
Met watter hoek lei v_1 vir v_2 indien $v_1(t) = -4\sin(\omega t+10°)V$ en $v_2(t) = 5\cos(\omega t-40°)V$ A) 65° B) 155° C) 335° D) 245° E) Geen van die bogenoemdes nie.	Find the angle v_1 leads v_2 by if $v_1(t) = -4\sin(\omega t+25°)V$ and $v_2(t) = 5\cos(\omega t-40°)V$. A) 65° B) 155° C) 335° D) 245° E) None of the above

Vraag 1.23	Question 1.23
Watter een van die volgende stellings in verband met fasors is vals? A) As 'n sinusodale funksie gedifferensieer word is dit dieselfde as om sy fasor met $j\omega$ te vermenigvuldig. B) As 'n sinusodale funksie geintegreer word is dit dieselfde as om sy fasor met $j\omega$ te deel. C) As twee sinusodale funksies opgetel word is dit dieselfde as om hulle fasors op te tel. D) 'n Fasor is 'n komplekse getal wat die amplitude en frekwensie van 'n sinusodale funksie aandui. E) Geen van die bogenoemdes nie.	Which one of the statements regarding phasors is false? A) Differentiating a sinusoid is equivalent to multiplying its corresponding phasors by $j\omega$ B) Integrating a sinusoid is equivalent to dividing its corresponding phasors by $j\omega$ C) Adding sinusoids of the same frequency is equivalent to adding their corresponding phasors. D) A phasor is a complex number that represents the amplitude and frequency of a sinusoid E) None of the above

Vraag 1.24	Question 1.24
Vraag 1.24 verwys na Fig. Q1.24. Bepaal die waarde van die spanning oor die 1Ω resistor by 100Hz. A) 9.99V B) 8.65V C) 3.78V D) 9.95V E) Geen van die bogenoemdes nie.	Question 1.24 refers to Fig. Q1.24. Find the magnitude of the voltage across the 1Ω resistor at 100Hz. A) 9.99V B) 8.65V C) 3.78V D) 9.95V E) None of the above

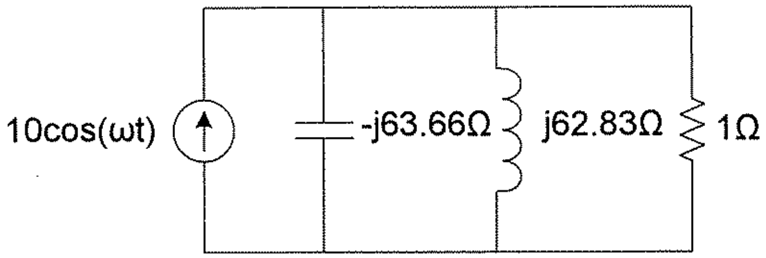


Fig. Q1.24

Vraag 1.25	Question 1.25
Vraag 1.25 verwys na Fig. Q1.25. Bepaal die totale impedansie Z_T. Aanvaar ω is 6280 rad/s. A) (50+j0.89)Ω B) (50-j0.89) Ω C) 50 Ω D) (50+j62.8) Ω E) Geen van die bogenoemdes nie.	Question 1.25 refers to Fig. Q1.25. Find the total impedance Z_T. Assume ω is 6280 rad/s. A) (50+j0.89)Ω B) (50-j0.89) Ω C) 50 Ω D) (50+j62.8) Ω E) None of the above

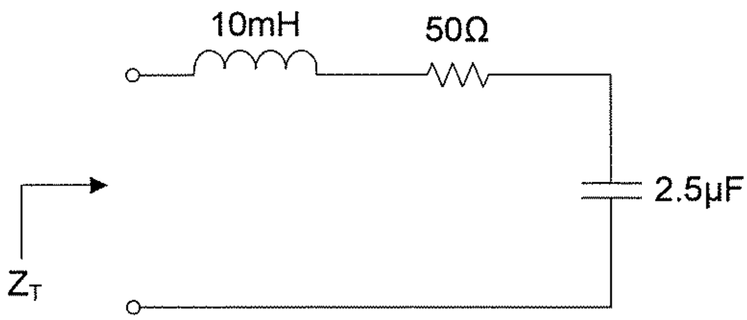


Fig. Q1.25

Vraag 1.26	**Question 1.26**
'n 100µF capacitor en 'n 50Ω resistor is in parallel. Indien die totale reaktansie nul moet wees, watter waarde inductor moet in serie met die RC-baan geplaas word by 1kHz? A) 255µH B) 113µH C) 25.5µH D) 1.13mH E) Geen van die bogenoemdes nie.	A 100µF capacitor and a 50Ω resistor are in parallel. What value of inductor should be placed in series with the RC-circuit to yield zero reactance at 1kHz? A) 255µH B) 113µH C) 25.5µH D) 1.13mH E) None of the above

Vraag 1.27	**Question 1.27**
Indien drie strome in 'n node invloei in 'n stroombaan, vind $i_1(t)$ as $I_2 = 25\angle 120°$ mA en $I_3 = 35\angle 25°$ mA by 60Hz. A) -60cos(377t-95°)mA B) 20.4cos(377t-160.4°)mA C) 20.4cos(377t-19.6°)mA D) 60sin(377t-95°)mA E) Geen van die bogenoemdes nie.	If three currents enter a node in a circuit, find $i_1(t)$ if $I_2 = 25\angle 120°$ mA and $I_3 = 35\angle 25°$ mA at 60Hz. A) -60cos(377t-95°)mA B) 20.4cos(377t-160.4°)mA C) 20.4cos(377t-19.6°)mA D) 60sin(377t-95°)mA E) None of the above

Vraag 1.28	**Question 1.28**
Die stroom deur 'n 10mH induktor word gegee as $\mathbf{I} = 34.5\angle 12°$ A. Vind die spanning oor die induktor by 314.2 rad/s. A) $108.4\angle 102°$ B) $96.5\angle 102°$ C) $96.5\angle -78°$ D) $108.4\angle -78°$ E) Geen van die bogenoemdes nie.	Let the current into a 10mH inductor be given by $\mathbf{I} = 34.5\angle 12°$ A. Find the voltage across the inductor at 314.2 rad/s. A) $108.4\angle 102°$ B) $96.5\angle 102°$ C) $96.5\angle -78°$ D) $108.4\angle -78°$ E) None of the above

Vraag 1.29	**Question 1.29**
Vraag 1.29 verwys na Fig. Q1.29. Bepaal die waarde van R_L wat maksimum drywings oordrag na die las verseker, en die frekwensie waarby dit plaasvind. A) 10Ω, 500Hz B) 10Ω, 1kHz C) 5Ω, 1kHz D) 5Ω, 500Hz E) Geen van die bogenoemdes nie.	Question 1.29 refers to Fig. Q1.29. Determine the value of R_L for maximum power transfer to the load and the frequency at which it occurs. A) 10Ω, 500Hz B) 10Ω, 1kHz C) 5Ω, 1kHz D) 5Ω, 500Hz E) None of the above

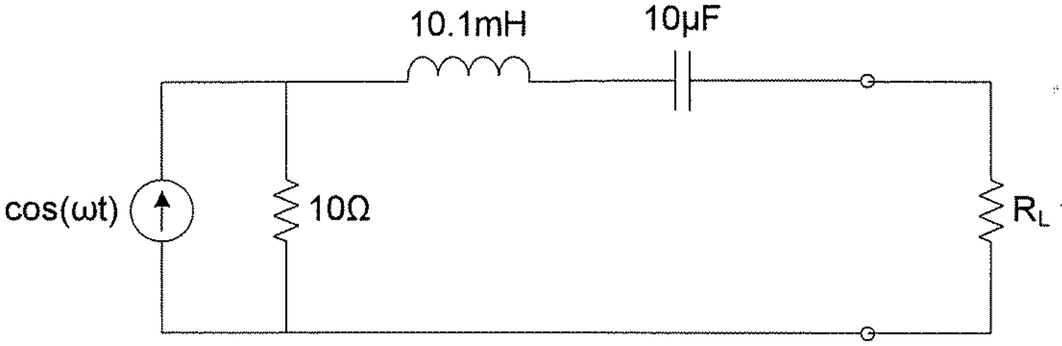


Fig. Q1.29

Vraag 1.30 Vraag 1.30 verwys na Fig. Q1.30. Bepaal die Norton ekwivalnete R_N vir die stroombaan. A) $0.45\angle{-21.26}\ \Omega$ B) $0.45\angle{-2.126}\ \Omega$ C) $1000\angle{-21.26}\ \Omega$ D) $1000\angle{-2.126}\ \Omega$ E) Geen van die bogenoemdes nie.	**Question 1.30** Question 1.30 refers to Fig. Q1.30. Determine the Norton equivalent R_N for the circuit. A) $0.45\angle{-21.26}\ \Omega$ B) $0.45\angle{-2.126}\ \Omega$ C) $1000\angle{-21.26}\ \Omega$ D) $1000\angle{-2.126}\ \Omega$ E) None of the above

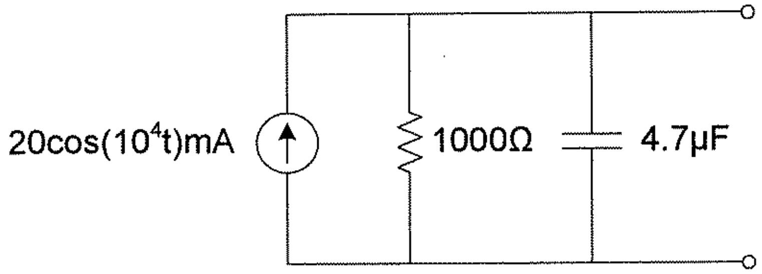
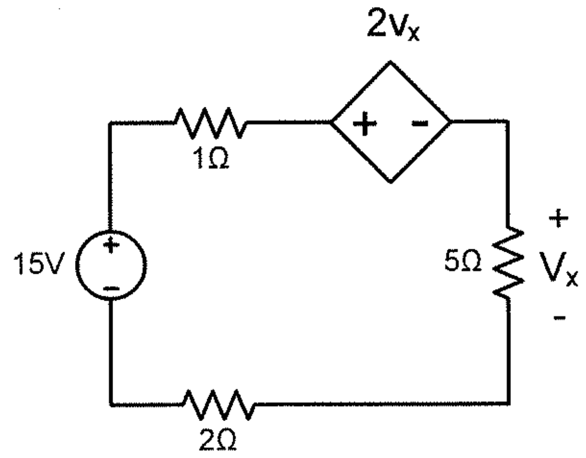


Fig. Q1.30

VRAAG 2 / QUESTION 2

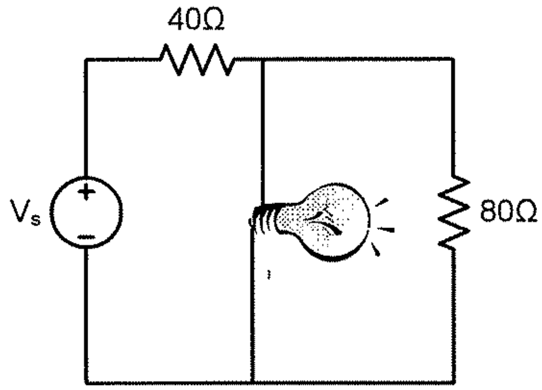
Vraag 2.1	Question 2.1



Vind v_x in die stroombaan.	Find v_x in the circuit.

[2]

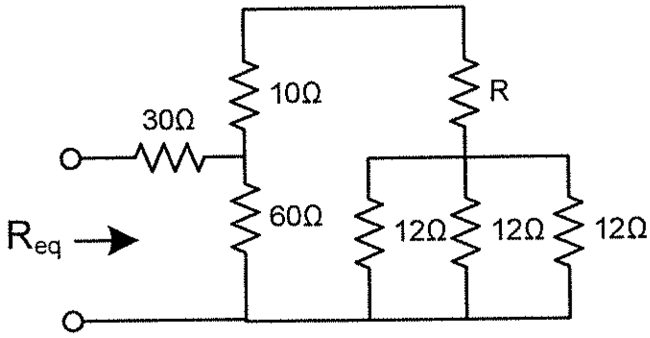
Vraag 2.2	Question 2.2



Die gloeilamp het 'n spesifikasie van 220V, 0.75A. Vind V_s sodat die gloeilamp werk by sy spesifikasiewaardes.	The lightbulb is rated 220V, 0.75A. Find V_s to make the lightbulb operate at its rated condition.

[2]

Vraag 2.3	Question 2.3

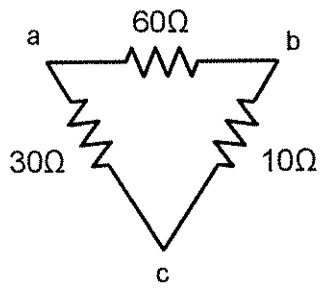


As R_{eq} = 50Ω, bepaal R.

If R_{eq} = 50Ω, determine R.

[2]

Vraag 2.4	Question 2.4

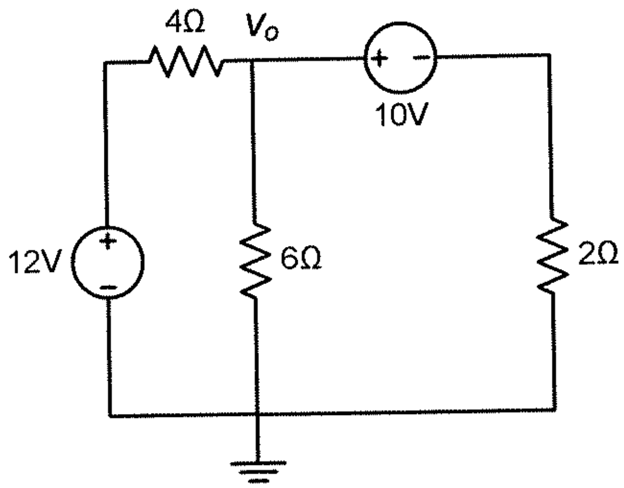


Skakel die stroombaan om van Δ na Y.

Convert the circuit from Δ to Y.

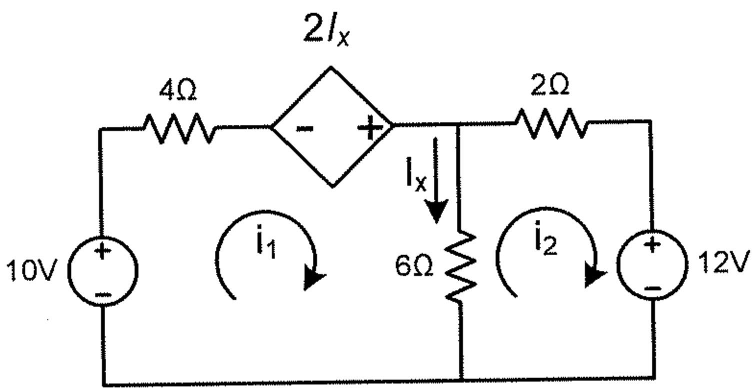
[3]

Vraag 2.5	Question 2.5



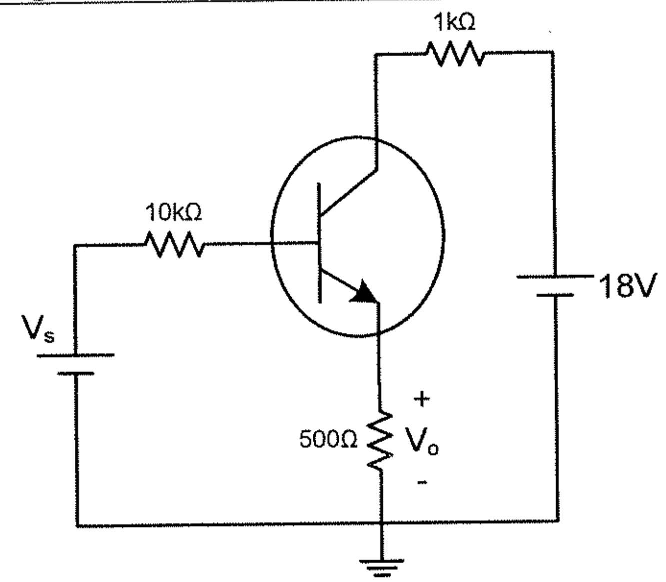
Gebruik NODALE analise on v_o te bepaal.	Use NODAL analysis to determine v_o. **[2]**

Vraag 2.6	Question 2.6



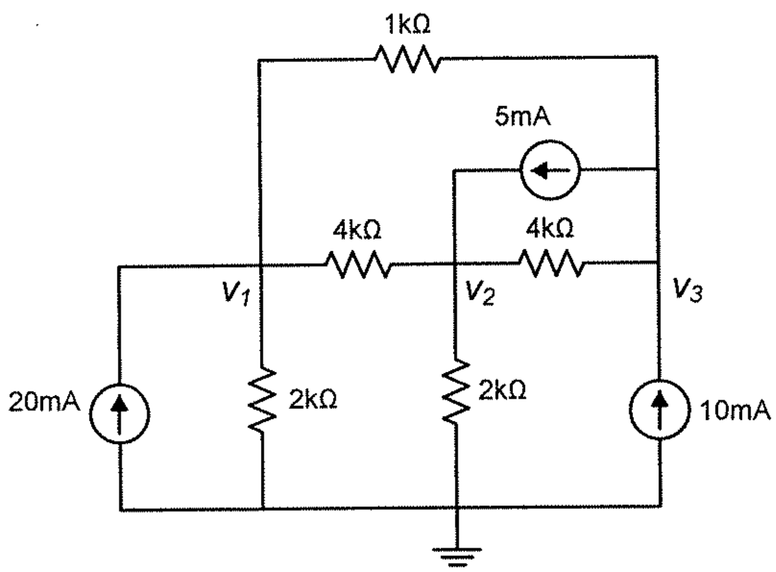
Bepaal die lusstrome i_1 en i_2.	Determine the mesh currents i_1 and i_2. **[4]**

Vraag 2.7	Question 2.7



Bepaal V_s vir die transistorbaan as V_o=4V, β=150 en V_{BE}=0.7V.	Determine V_s for the transistor circuit if V_o=4V, β=150 and V_{BE}=0.7V. **[2]**

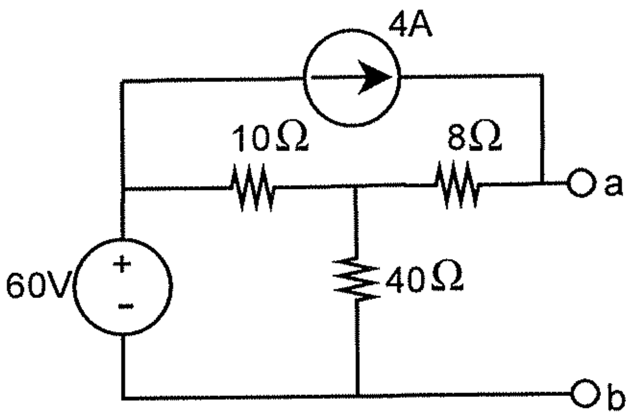
Vraag 2.8	Question 2.8



Bepaal die node-spanningsvergelykings deur inspeksie. Neem alle konduktansies in **mS** en alle strome in **mA**. Verskaf die **G** matriks.	Determine the node-voltage equations by inspection. Assume all conductances in **mS** and all currents in **mA**. Provide the **G** matrix. **[3]**

VRAAG 3 / QUESTION 3

Vraag 3.1	Question 3.1

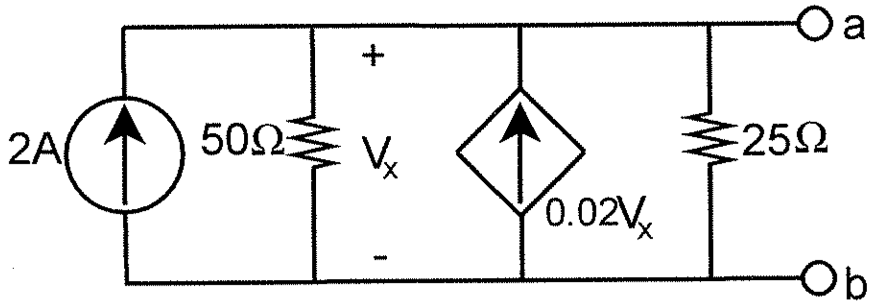


(a) Bepaal R_{th} met verwysing tot terminale a–b.
(b) Bepaal die bydrae van die 4A stroombron tot die ope-baan spanning V_{ab}.
(c) Bepaal die bydrae van die 60V spannings-bron tot die ope-baan spanning V_{ab}.

(a) Find R_{th} with respect to terminals a–b.
(b) Find the contribution of the 4A current source to the open-circuit voltage V_{ab}.
(c) Find the contribution of the 60V voltage source to the open-circuit voltage V_{ab}.

[6]

Vraag 3.2	Question 3.2

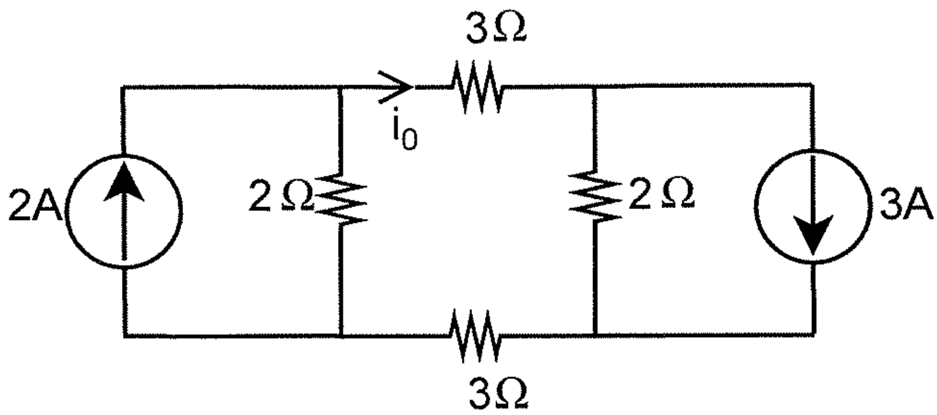


Bepaal die Norton-ekwivalent met verwysing tot terminale a-b.

Determine the Norton equivalent with respect to terminals a-b.

[4]

Vraag 3.3	Question 3.3

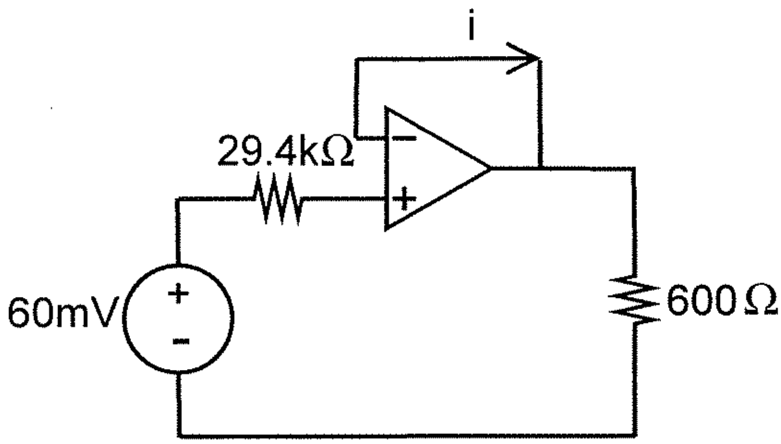


Gebruik brontranformasies om die stroom i_0 te bepaal in die onderstaande stroombaan.

Use source transformations to find the current i_0 in the circuit below.

[2]

Vraag 3.4	Question 3.4

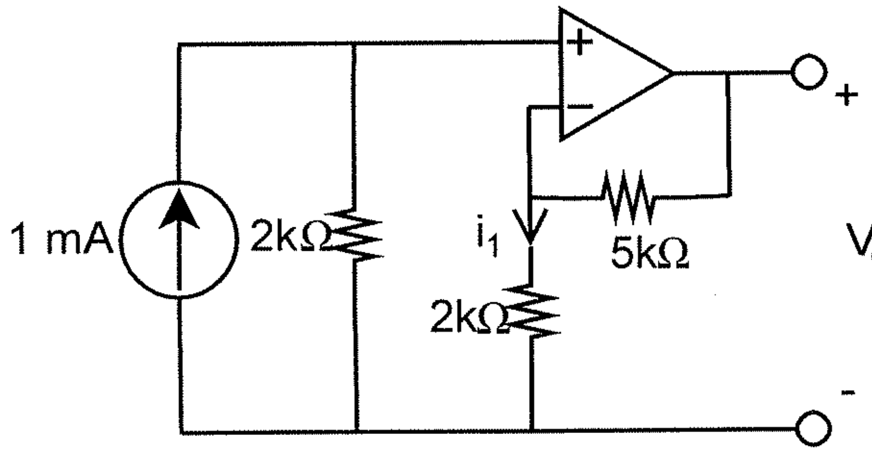


Aanvaar 'n ideale operasionele versterker. Bereken die stroom *i* en die drywing gedissipeer deur die 600 Ω weerstand.

Assume an ideal op-amp. Calculate the current *i* and the power absorbed by the 600 Ω resistor.

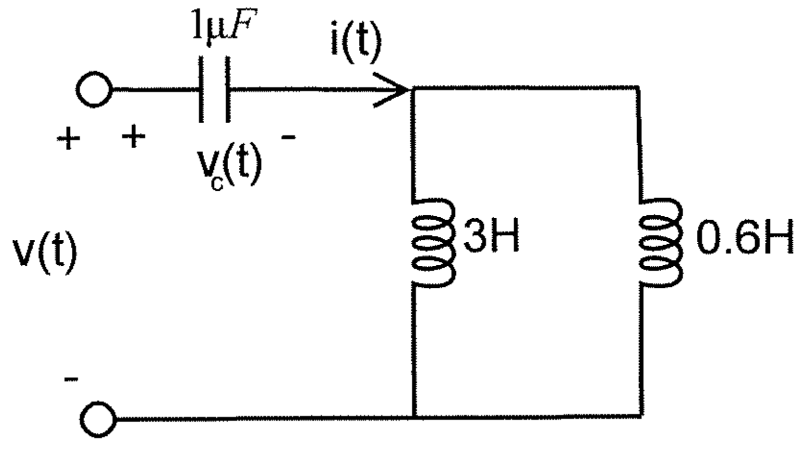
[2]

Vraag 3.5	Question 3.5



Aanvaar 'n ideale operasionele versterker. Bepaal i_1 and V_0 in die stroombaan.	Assume an ideal op-amp. Determine i_1 and V_0 in the circuit. **[3]**

Vraag 3.6	Question 3.6



Indien die spanning oor die kapasitor gegee word as $v_c(t) = 4\cos(2000t)$, bepaal i(t), die stroom deur die kapasitor, asook v(t), die totale spanning oor die inset-terminale.	If the voltage across the capacitor is given as $v_c(t) = 4\cos(2000t)$, determine i(t), the current through the capacitor, as well as v(t), the total voltage at the input terminals. **[3]**

VRAAG 4 / QUESTION 4

Vraag 4.1	Question 4.1

$$\text{(a)}\ \frac{15\angle 45^{\circ}}{3-j4}+j2$$

$$\text{(b)}\ \frac{8\angle -20^{\circ}}{(2+j)(3-j4)}$$

Bereken die bg. komplekse getalle en druk die antwoord in reghoekige vorm uit.	Calculate the above complex numbers and express your results in rectangular form. **[2]**

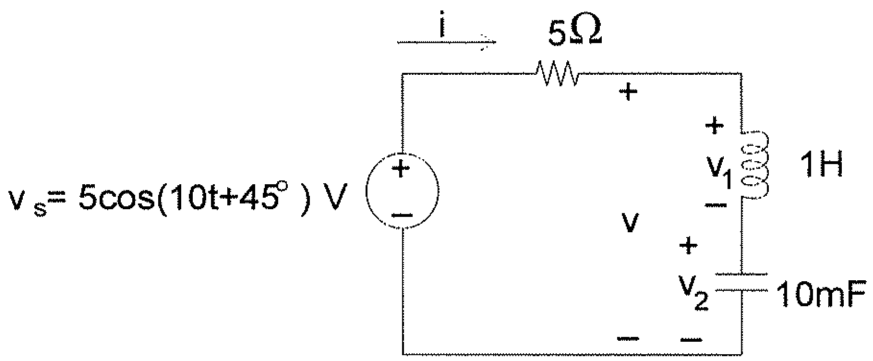
Vraag 4.2	Question 4.2

(a) $v = 20\cos(\omega t + 25^{\circ}) - 10\cos(\omega t - 90^{\circ})$V

(b) $i = 25\cos(\omega t - 45^{\circ}) - 10\sin(\omega t + 45^{\circ})$A

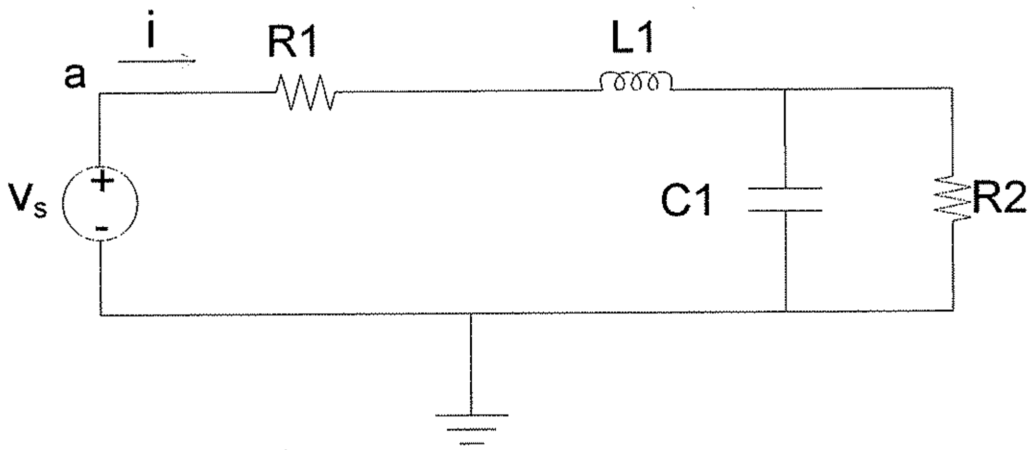
Pas fasoranalise toe om v en i in sinusoidale vorm te bepaal.	Apply phasor analysis to determine v and i in sinusoidal form. **[2]**

Vraag 4.3	Question 4.3



Gebruik fasoranalise en bepaal i(t), v(t), $v_1(t)$ en $v_2(t)$.	Use phasor analysis to determine i(t), v(t), $v_1(t)$ and $v_2(t)$. **[4]**

Vraag 4.4	Question 4.4



a) Bepaal die ekwivalente impedansie wat die bron tussen tussen nodus a en grond sien, as ω = 200 rad/s en R1 = 4kΩ, R2 = 1kΩ, L1 = 25mH en C1 = 10μF.

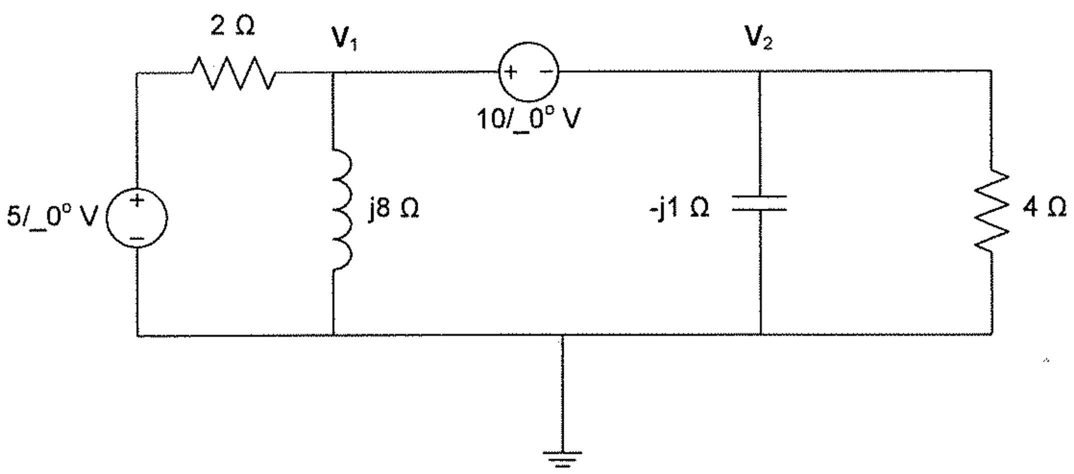
b) Aanvaar die ekwivalente impedansie tussen nodus a en grond is (1500 – *j*150) Ω. Bepaal die stroom i indien v_s = 60cos(200*t* –10°) V.

a) Determine the equivalent impedance seen by the source between node a and ground, if ω = 200 rad/s and R1 = 4kΩ, R2 = 1kΩ, L1 = 25mH en C1 = 10μF.

b) Assume the equivalent impedance between node a and ground is (1500 - *j*150) Ω. Determine the current i if v_s = 60cos(200*t* –10°) V.

[4]

Vraag 4.5	Question 4.5

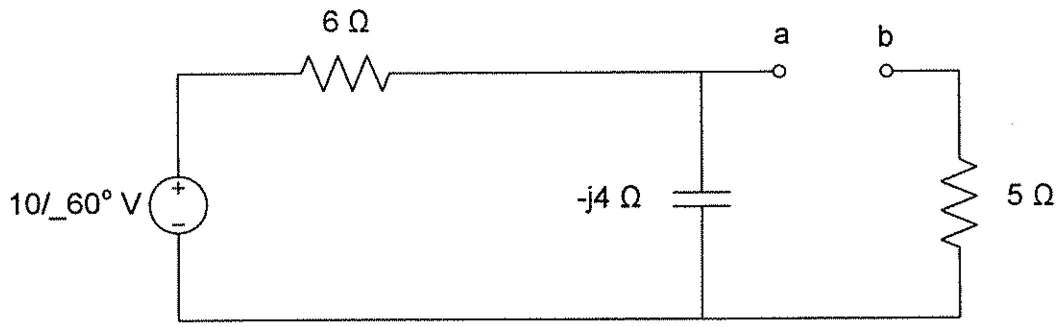


Gebruik nodale analise om die fasordomein nodespannings V_1 en V_2 te bepaal.

Use nodal analysis to determine the phasor domain node voltages V_1 and V_2.

[4]

Vraag 4.6	Question 4.6



Bepaal die **Norton** ekwivalente baan van die bg. stroombaan.	Determine the **Norton** equivalent of the above circuit. **[4]**

DIE EINDE / THE END